IPR & TT CELL
VIT VELLORE

VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

ENTREPRENEURSHIP DEVELOPMENT CELL
VIT VELLORE

TECHNOLOGY BRIEF

Technology ID: 2024VITTTO73

Natural Vitamin B12 Fortification in Fermented Foods

Technology Domain

- Food Biotechnology
- Fermentation Microbiology
- Nutraceutical Engineering

Applications

- Staple Batter Enrichment
- Bakery Products
- Dairy Fermentation

Problem Statement

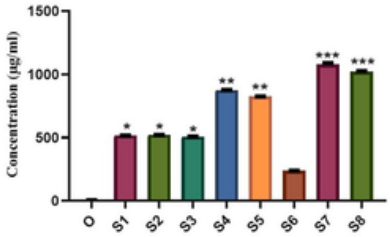
- Plant-based diets lack sufficient Vitamin B12
- Significant loss of Vitamin B12 during cooking
- Critical deficiency in vegetarians and vegans, particularly in pregnant women and nursing mothers
- Vitamin B12 deficiency leads to neurological disorders, pernicious anemia, cardiovascular risks, mental health risks, GI disorders, etc.

Technological Solution

- Formulation comprising inexpensive additives with food products before fermentation
- Synergistic enhancement of Vitamin B12 level directly within food matrix during its natural fermentation process
- Novel approach ensuring Vit B12 availability in food avoiding need for supplements

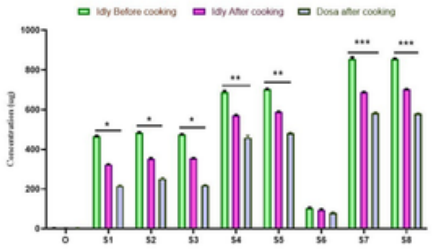
Results

Synergistic Enhancement of Vitamin B12 in curdled milk





Synergistic Enhancement of Vitamin B12 in Idly and Dosa batter





Technology Summary

This novel composition and method comprises mixing of specific amino acids (1 mg/ml Glycine, 1 mg/ml K-glutamate or Mono-sodium glutamate) and 1 µg/ml Cobalt(II) chloride with a food product (e.g., yogurt, idli batter, bread dough) before fermentation.

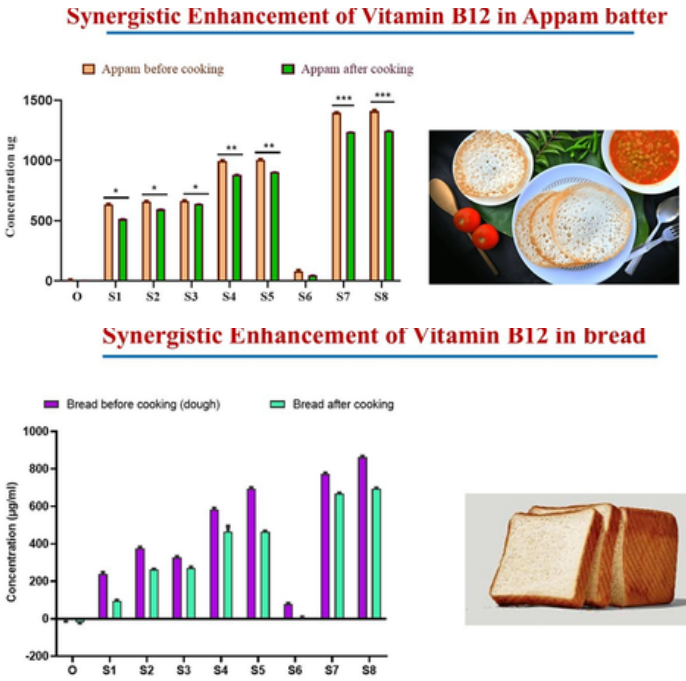
This technology establishes a remarkable increase in vitamin B12 concentration in various fermented foods (e.g., up to 1409 µg/ml in appam batter, 1025 µg/ml in curdled milk), with minimal reduction (only 20%) after cooking, indicating stability.

Technology available for Licensing

Technical USP

- Simple integration into fermented food using minimal process tweaks and budget-friendly additives.
- Boosts bioavailability naturally, making superfoods accessible, scalable, and profitable for producers everywhere.

Results



Competitive Advantage

- Superior bioavailability of Vitamin B12 when paired with food vs. pure supplements or injections
- Cost-effective alternative to pricey supplements or unavailable fortified foods, accessible to all societal sections
- Versatile for all ages: pregnant/lactating women, growing children, and aging populations with lifestyle diseases like diabetes and CVD

IP & Development Stage

- Patent Application: 202441015300 (Priority Date: March 1, 2024) - Patent Pending
- Technology Readiness Level - 5
- Ready for Industry scale-up

Value Proposition

- Ultra-Cost-Effective B12 Boost: Enrich essential Vitamin B12 directly in food—no premium ingredients needed
- Post-cooking B12 Retention: 80% of Vitamin B12 is retained in the food even after cooking (steaming, frying, or baking)
- Seamless Integration: Mix additives in-situ during fermentation or pre-ferment blending
- Ready-to-Sell Sachets: Package B12-enriching pre-mixes for instant scalability

Market Potential

- Global Vitamin B12 market: USD 350M (2024), reaching USD 593M by 2032 (6.8% CAGR)
- Indian market: USD 7-12M, growing to USD 20M+ by 2033 (6.4% CAGR), with 47% deficiency among 574M vegetarians
- Cost-effective, stable B12 method (1400+ µg/ml in batter) targets billions of vegetarians via sachet sales, premium fortified foods, and emerging-market Vitamin B12 deficiency

Research Lab

- Dr. Krishnan Venkataraman
  Professor, Centre for Bio Separation Technology (CBST), VIT Vellore

Technology available for Licensing

Contact:
Mr. Manish Kumar
Sr. Technology Transfer Officer
IP&TT Cell, VIT-IST, VIT Vellore

Email: manish.k@vit.ac.in
Phone No.: +91 416 220 4330 (O)
: +91 83099 93511 (M)
Website: https://vit.ac.in/iprcell